Kernix 0.7.3 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

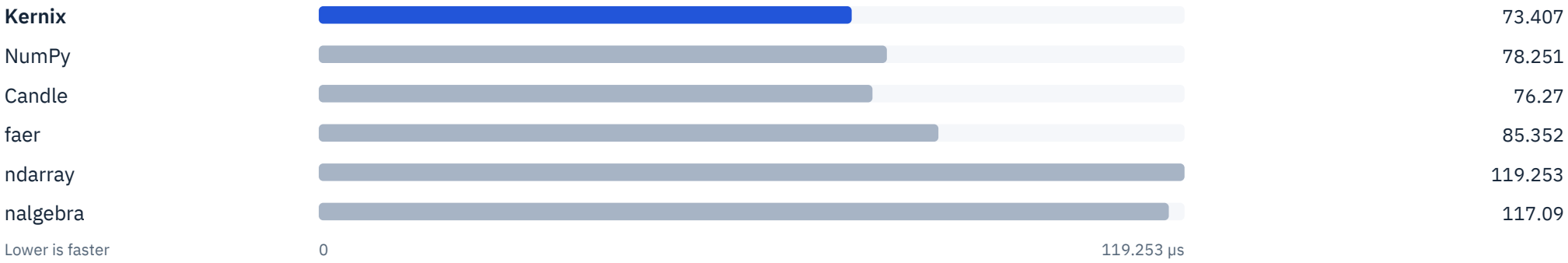
21 workloads

88 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

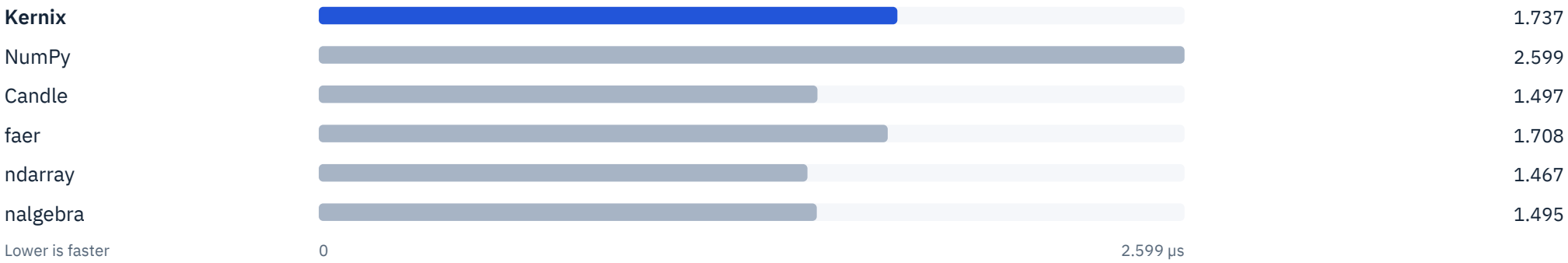
Kernix 0.7.3 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

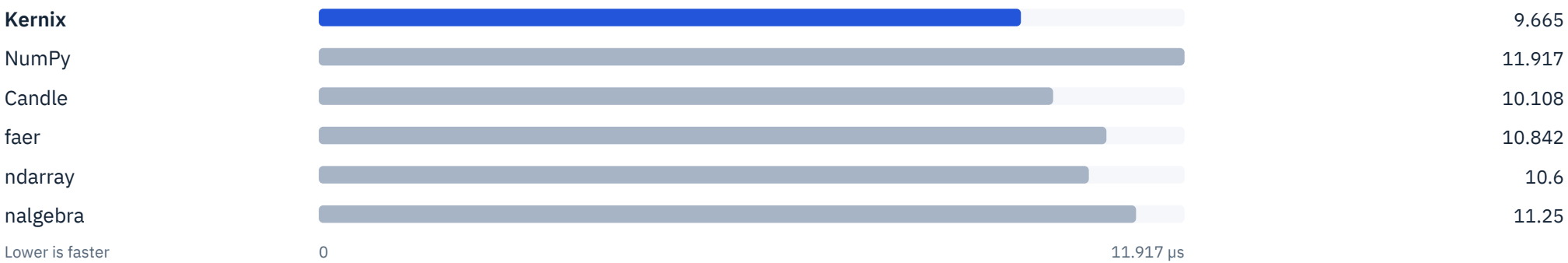
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



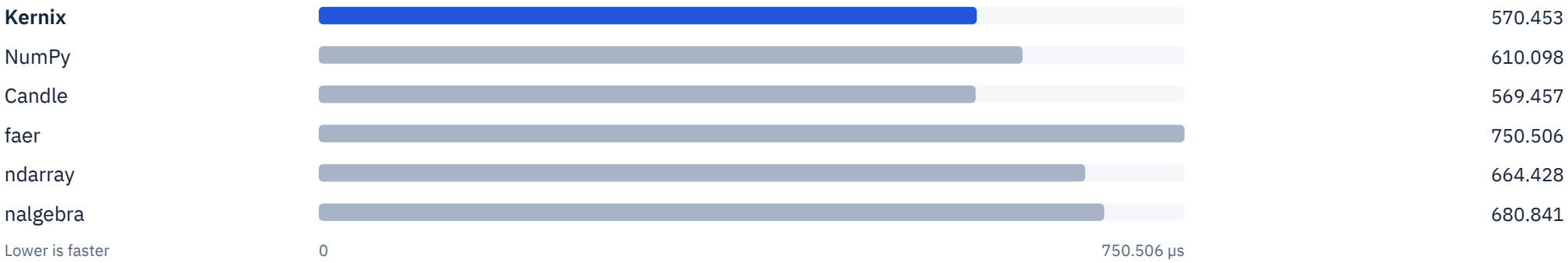
Kernix 0.7.3 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

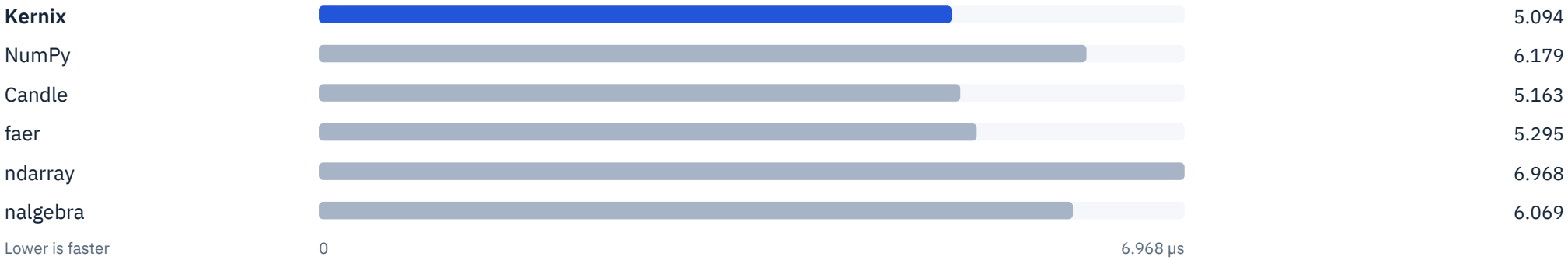
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



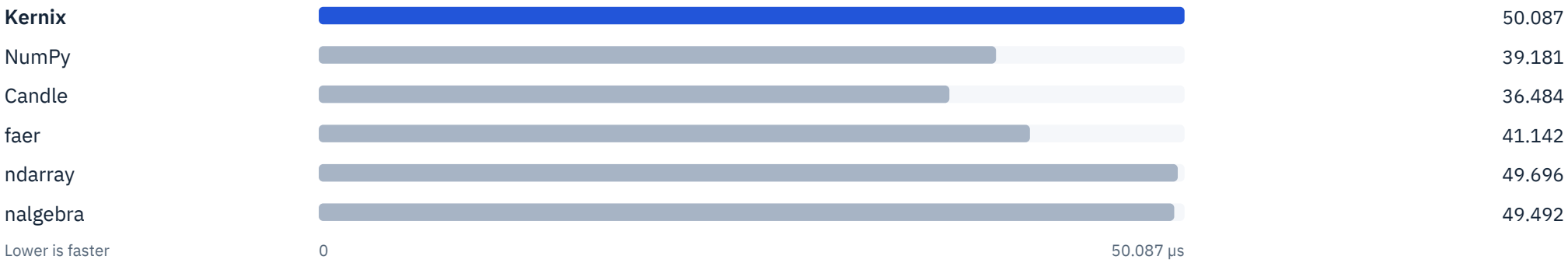
Kernix 0.7.3 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

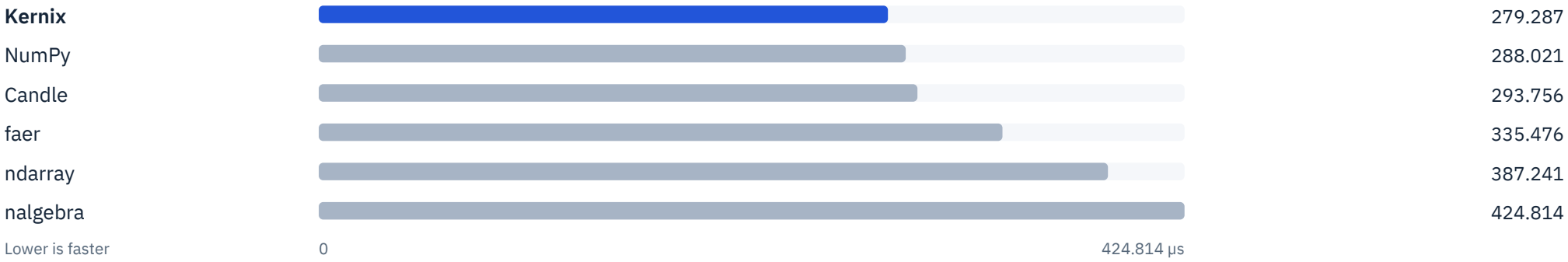
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



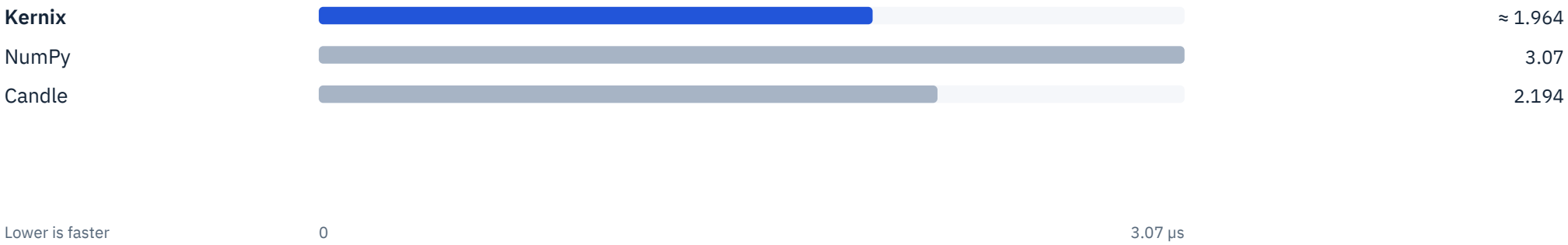
Kernix 0.7.3 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

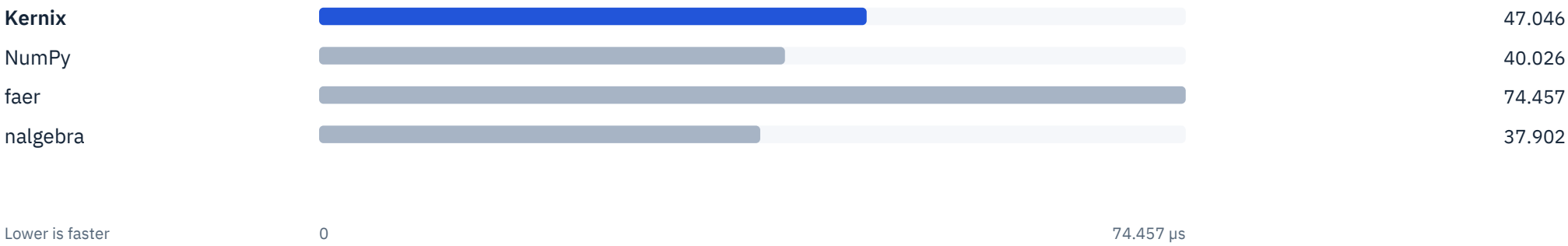


Kernix 0.7.3 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

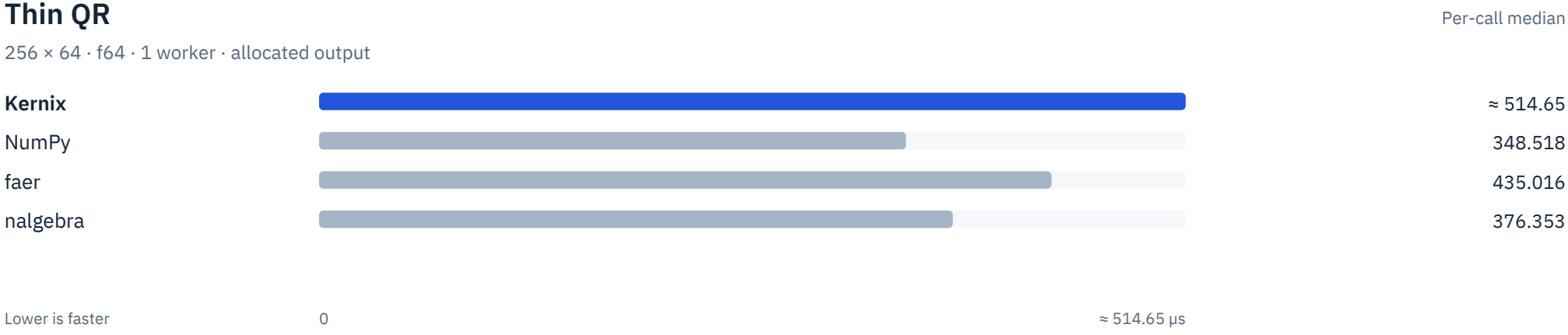
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



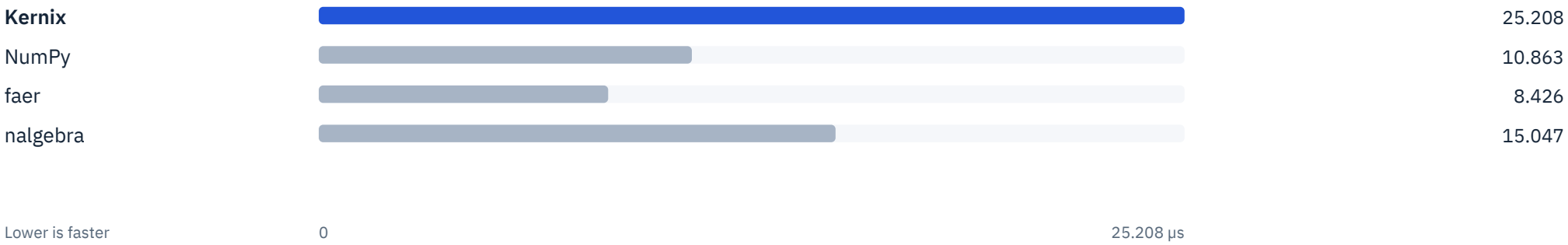
Kernix 0.7.3 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

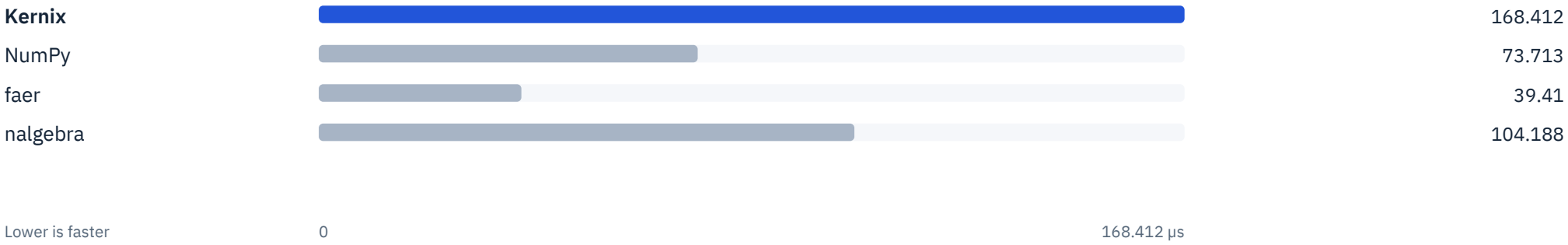
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



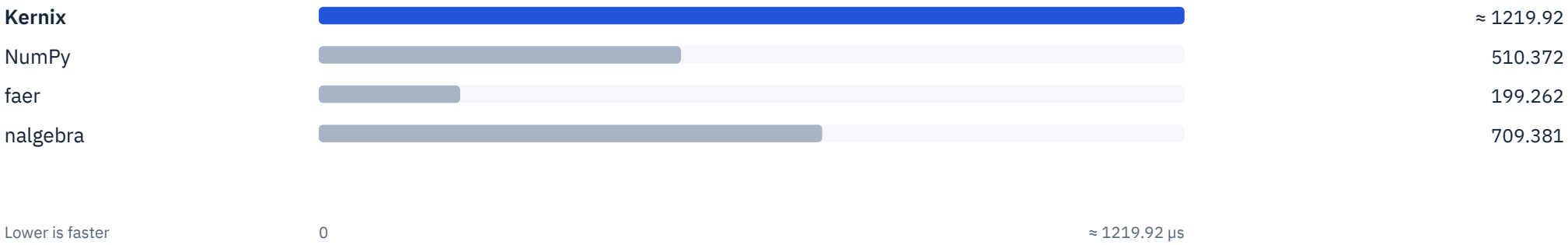
Kernix 0.7.3 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.7.3 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

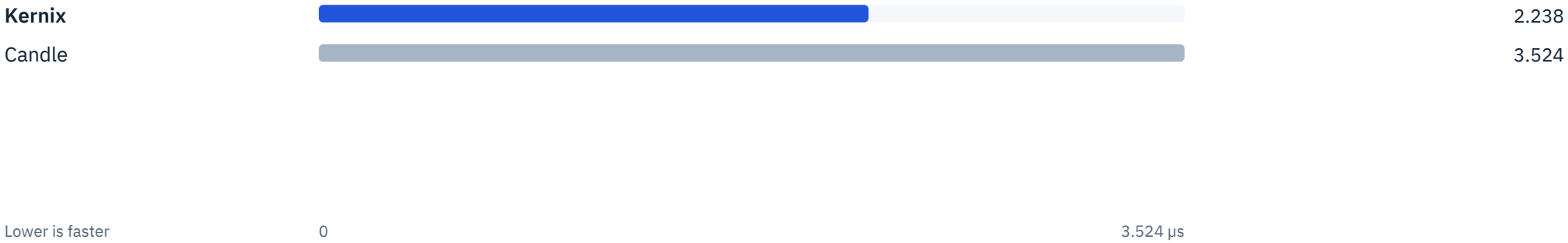
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.7.3 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

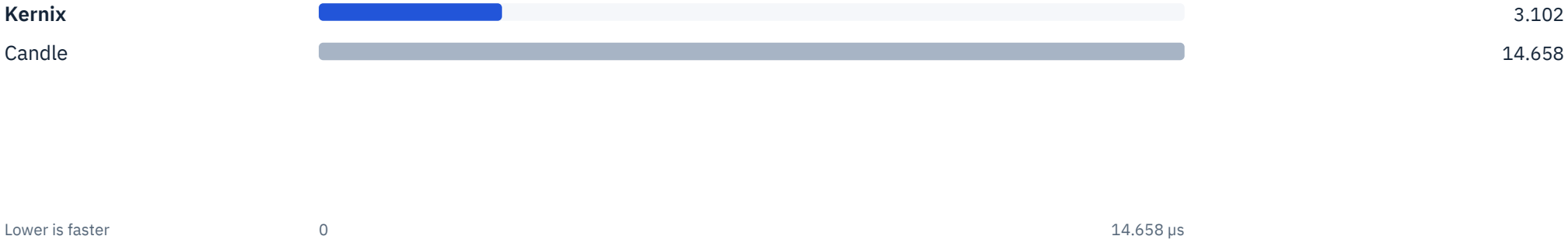
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.7.3 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

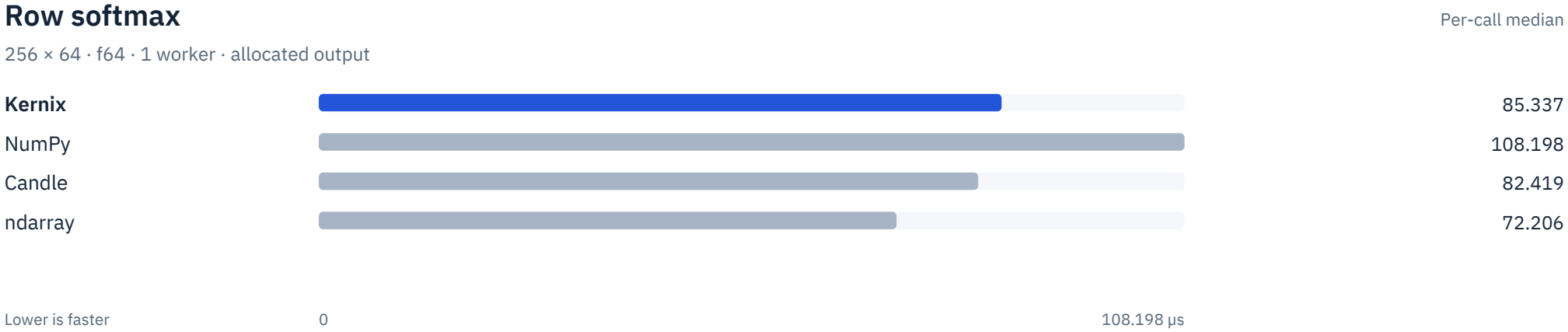


Kernix 0.7.3 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



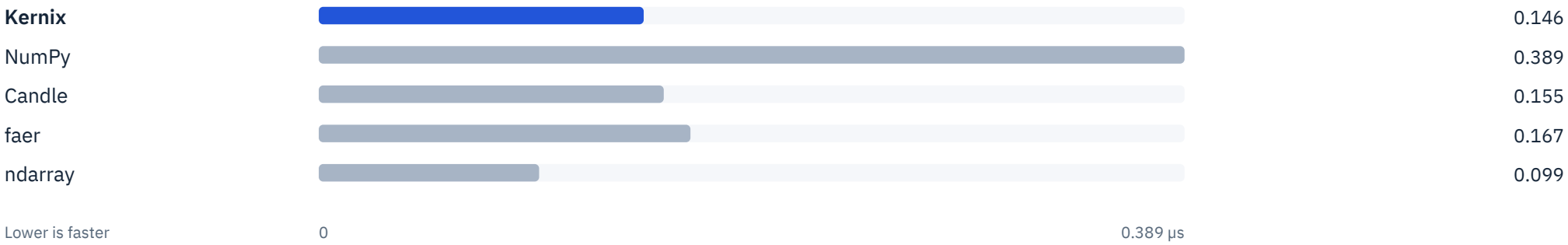
Kernix 0.7.3 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1737	1.737
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1467	1.467
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1708	1.708
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1497	1.497
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2599	2.599
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1495	1.495
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11250	11.25
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	11917	11.917
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9665	9.665
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	10600	10.6

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	10842	10.842
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	10108	10.108
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	117090	117.09
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	85352	85.352
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	119253	119.253
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	73407	73.407
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	78251	78.251
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	76270	76.27
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	680841	680.841
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	750506	750.506

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	610098	610.098
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	570453	570.453
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	569457	569.457
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	664428	664.428
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6069	6.069
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	5163	5.163
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	5094	5.094
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6179	6.179
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5295	5.295
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6968	6.968

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	49696	49.696
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	39181	39.181
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	49492	49.492
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	50087	50.087
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	36484	36.484
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	41142	41.142
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	335476	335.476
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	293756	293.756
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	424814	424.814
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	279287	279.287

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	288021	288.021
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	387241	387.241
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2194	2.194
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3070	3.07
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1964.0000000000002	1.9640000000000002
Thin QR / Kernix 96 × 32 · f64	47046	47.046
Thin QR / faer 96 × 32 · f64	74457	74.457
Thin QR / nalgebra 96 × 32 · f64	37902	37.902
Thin QR / NumPy 96 × 32 · f64	40026	40.026
Thin QR / Kernix 256 × 64 · f64	514649.99999999994	514.64999999999994

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 256 × 64 · f64	435016	435.016
Thin QR / nalgebra 256 × 64 · f64	376353	376.353
Thin QR / NumPy 256 × 64 · f64	348518	348.518
Cholesky factorization / nalgebra 64 × 64 · f64	15047	15.047
Cholesky factorization / Kernix 64 × 64 · f64	25208	25.208
Cholesky factorization / faer 64 × 64 · f64	8426	8.426
Cholesky factorization / NumPy 64 × 64 · f64	10863	10.863
Cholesky factorization / NumPy 128 × 128 · f64	73713	73.713
Cholesky factorization / nalgebra 128 × 128 · f64	104188	104.188
Cholesky factorization / faer 128 × 128 · f64	39410	39.41

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	168412	168.412
Cholesky factorization / faer 256 × 256 · f64	199262	199.262
Cholesky factorization / nalgebra 256 × 256 · f64	709381	709.381
Cholesky factorization / Kernix 256 × 256 · f64	1219916	1219.916
Cholesky factorization / NumPy 256 × 256 · f64	510372	510.372
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2656	2.656
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	8284	8.284
Model inference / Candle batched normalized/sliced MLP · f64	3524	3.524
Model inference / Kernix batched normalized/sliced MLP · f64	2238	2.238
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2802	2.802

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Candle batched normalized/sliced MLP · f64	5153	5.153
Model backward / Candle prepared graph → fresh gradients · f64	14658	14.658
Model backward / Kernix prepared graph → fresh gradients · f64	3102	3.102
Parameter update / Kernix transactional SGD, fixed gradients · f64	532	0.532
Row softmax / NumPy 256 × 64 · f64	108198	108.198
Row softmax / Candle 256 × 64 · f64	82419	82.419
Row softmax / Kernix 256 × 64 · f64	85337	85.337
Row softmax / ndarray 256 × 64 · f64	72206	72.206
Create zeros / Kernix 64 × 64 · f32	146	0.146
Create zeros / faer 64 × 64 · f32	167	0.167

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / ndarray 64 × 64 · f32	99	0.099
Create zeros / NumPy 64 × 64 · f32	389	0.389
Create zeros / Candle 64 × 64 · f32	155	0.155
Create ones / faer 64 × 64 · f32	189	0.189
Create ones / Kernix 64 × 64 · f32	204	0.204
Create ones / Candle 64 × 64 · f32	230	0.23
Create ones / NumPy 64 × 64 · f32	1333	1.333
Create ones / ndarray 64 × 64 · f32	141	0.141